Enrolment Number										

Total No. of printed pages = 04

Winter, 2026

B.Tech 4th Semester Examination

Civil Engineering-Societal and Global Impact

Course Code: MNC-AC

Full Marks – 50 **Time – 2 hours**

The figure in the margin indicates full marks for the questions.

SECTION I

Answer all the questions **Marks: 10 x 1 = 10**

1(a) The core purpose of the civil engineering profession is to serve society by:

i) Prioritizing private sector profits for long-term sustainability
ii) Improving community infrastructure and quality of life
iii) Strictly enforcing administrative policies and government decisions
iv) Optimizing the cost of public construction projects

1(b) The primary objective of conducting a Life Cycle Assessment (LCA) of a building material is to evaluate:

i) The total financial cost from procurement to installation
ii) The environmental impact associated with all stages of a material's life, from raw extraction to disposal
iii) The structural load-bearing capacity over a fifty-year period
iv) The aesthetic compatibility with local architectural guidelines

1(c) Embodied energy in a construction material refers to:

i) The kinetic energy the material can absorb during seismic events
ii) The total energy consumed during raw material extraction, processing, manufacture, and transportation

iii) The electrical energy required to operate the completed facility

iv) The thermal energy retained by the material during peak sunlight hours

1(d) Which of the following critical national sub-sectors is categorized strictly as an "Economic Infrastructure" rather than a "Social Infrastructure"?

i) Public education facilities

ii) Institutional healthcare networks

iii) Power, irrigation, and transportation systems

iv) Low-income civic housing projects

1(e) In a smart city, which technology is most used for real-time traffic management?

i) GIS-based mapping

ii) IoT sensors and adaptive signal control

iii) Traffic policing

iv) CCTV only

1(f) Which project management paradigm relies on a strict, sequential linear progression where a phase must be fully completed and approved before the next phase begins?

i) Agile Project Management

ii) Lean Construction Paradigm

iii) Waterfall Paradigm

iv) Fast-Track Design-Build Matrix

1(g) In the Lean Construction framework, the primary operational focus is directed toward:

i) Consolidating the volume of material waste stored on site for use in future project phases

ii) Minimizing non-value-adding activities (waste) and optimizing entire workflow delivery through continuous collaboration

iii) Optimizing the structural safety factors below statutory municipal engineering limits

iv) Extending the total time duration of the planning and procurement lifecycle

1(h) Which of the following best demonstrates the concept of circular economy in construction?

i) Landfilling biodegradable demolition waste

ii) Reusing and recycling construction materials

iii) Importing new cost-effective materials and exporting finished products

iv) Burning waste in safe spaces

1(i) The best approach for retrofitting old buildings for energy efficiency is:

i) Adding insulation and upgrading windows

ii) Repainting walls in cooler colours

iii) Maximizing floor area

iv) Planting vegetation inside the building

1(j) Which of the following is a unique feature of the GRIHA rating system compared to LEED?

i) Focus on international building codes

ii) Emphasis on Indian climatic conditions and local construction practices

iii) Approved use of imported materials

iv) Exclusive assessment of energy efficiency

SECTION 2

Answer any one question from this section (300-500 words) **Marks: 10**

2(a) Explain the concept of life cycle assessment (LCA) and discuss its importance in sustainable civil engineering. (6+4)

2(b) Explain the concept of facility management and describe its role in addressing climate change. (6+4)

SECTION 3

Answer any one question from this section (300-500 words) **Marks: 10**

3(a) Demonstrate how the greenhouse effect systematically accelerates global warming and the broader climate change. What are the key strategies for mitigating GHG emissions in the context of civil engineering? (6+4)

3(b) Apply your knowledge of civil engineering techniques to model the construction of any famous building, monument or infrastructure of your choice, with reference to the specific challenges encountered during construction.

SECTION 4

Answer any one question from this section (300-500 words) **Marks: 10**

4(a) Analyze the basic concept and characteristics of infrastructure in the context of civil engineering. Distinguish between hard and soft infrastructure with examples. (6+4)

4(b) Examine the basic concept and techniques of sustainable construction with special reference to prefabrication and modular construction.

SECTION 5

Answer any one question from this section (300-500 words) **Marks: 10**

5(a) Critically assess the concept of sustainability and justify how civil engineering projects can contribute to sustainable development. (6+4)

5(b) Critically evaluate the extent to which the Smart City Mission projects in Guwahati have succeeded in improving civic infrastructure and quality of life with special reference to aesthetic projects like Brahmaputra Riverfront and persistent core issues like artificial flooding.